

The Room of Requirements for Developers

VOXXED DAYS CERN

FEBRUARY 2026

VOL I No I

Scaling AI Magic with Internal Portals



Maarten Vandepierre
Solutions Architect



Camille Nigon
Solutions Architect



Keep your options open



maarten-vandeperre.be

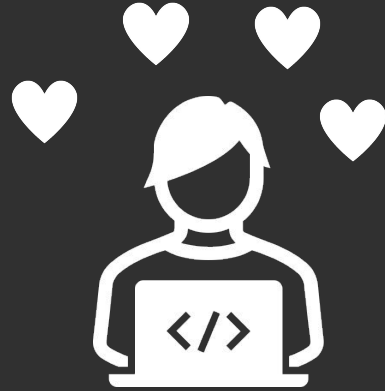


Muggle world

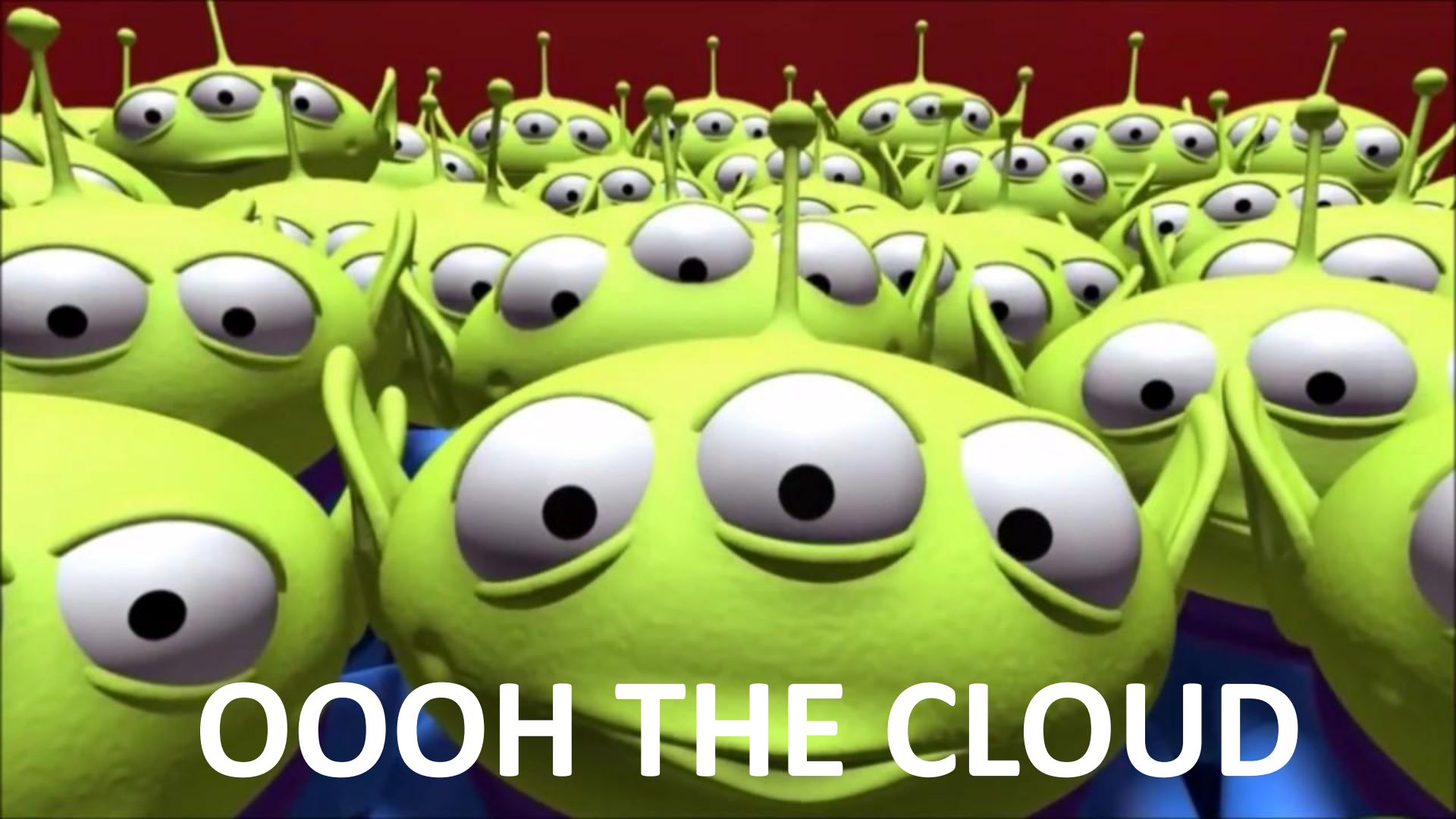
Platform engineering & AI-dev challenges



*Platform
Engineering?!*







OOOH THE CLOUD



$$\begin{aligned} H_0 &= H_1 - H_2 - H_3 - \dots \\ H_1 &= \sum_{i=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \\ H_2 &= \sum_{i=1}^{\infty} \langle \psi_i | \rho | \psi_i \rangle \\ &\vdots \\ H_3 &= \rho \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \dots \end{aligned}$$

$$G_7 - P_7 LS$$

What happened?!





*Shift Left:
Increasing
cognitive load*



It's not just
app development
anymore

Dev

Ops

DevOps

FinOps

MLOps

GitOps

DataOps

DevSecOps

BizDevOps

BizDevSecOps

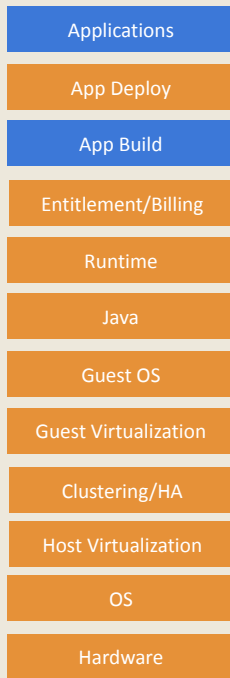
BizDevSecFinMLOps

Organizational Impact & Cognitive Load

Traditional App Delivery

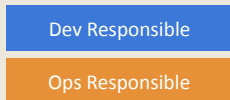


Dev



Ops

Key:

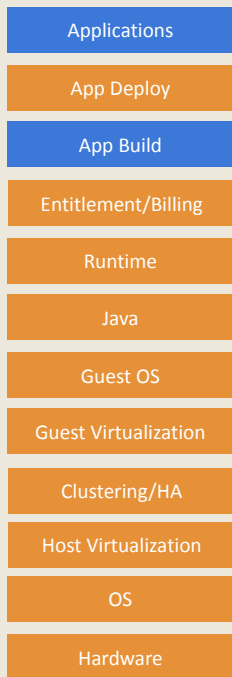


Organizational Impact & Cognitive Load

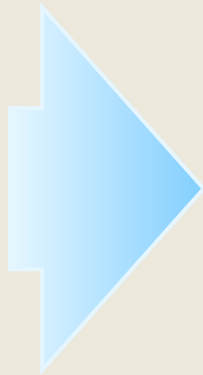
Traditional App Delivery



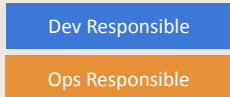
Dev



Ops



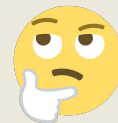
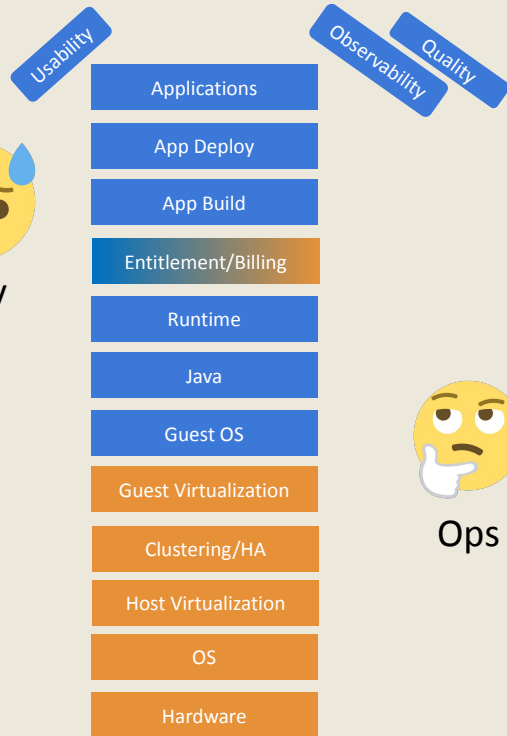
Key:



Modern App Delivery

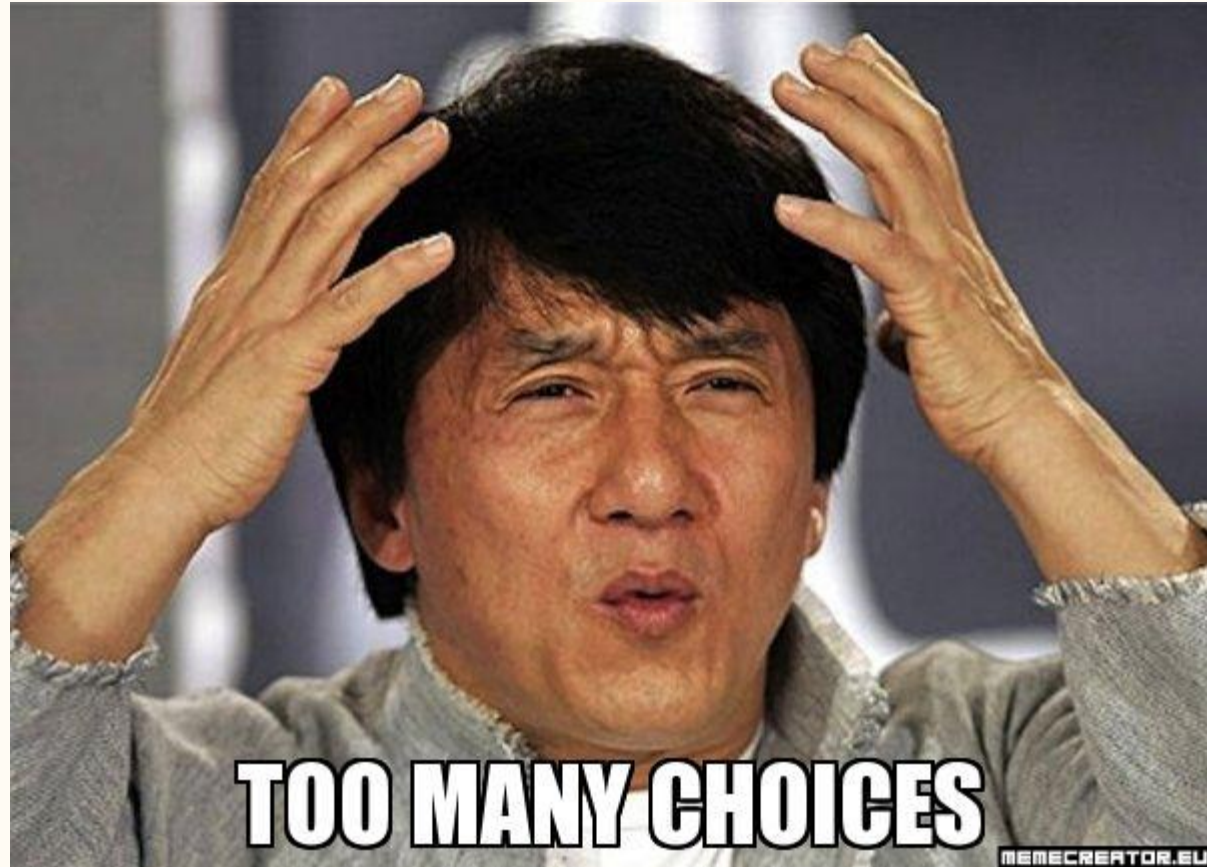


Dev



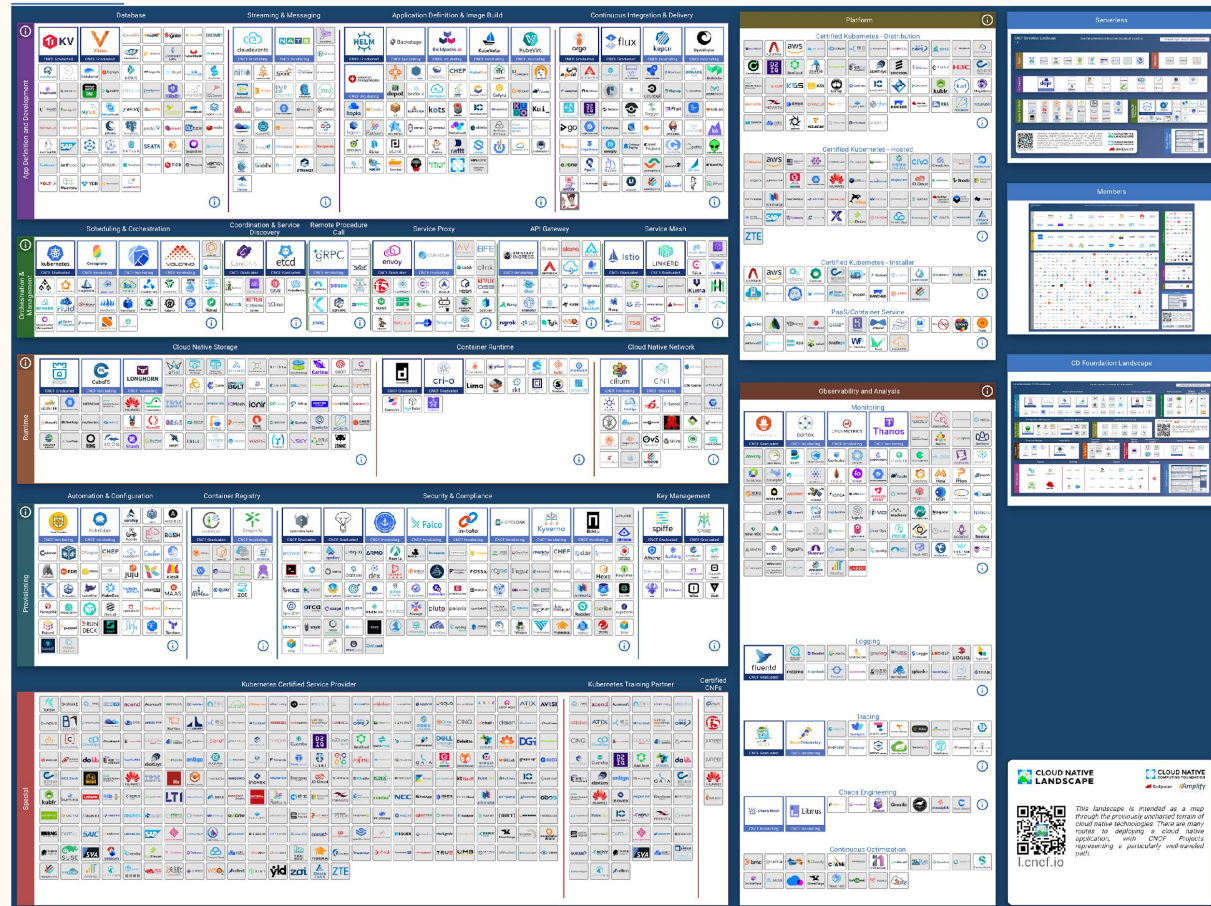
Ops

*Tools:
Too much
choice*



The Cloud-Native Grocery Store

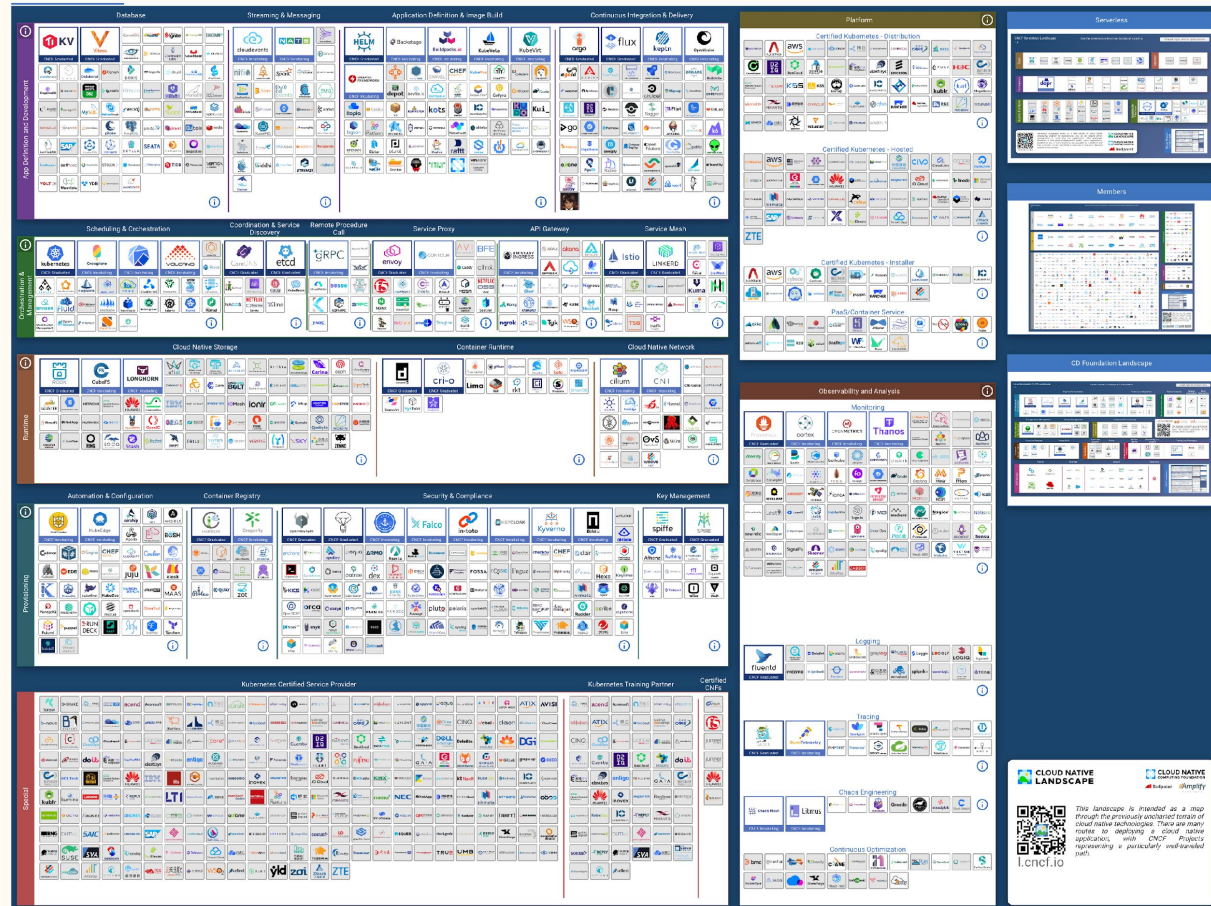
Where's Waldo?
Where's Charlie?
Where's Wally?



Source: <https://landscape.cncf.io/> - <https://nl.ign.com/harry-potter-the-series>

The Cloud-Native Grocery Store

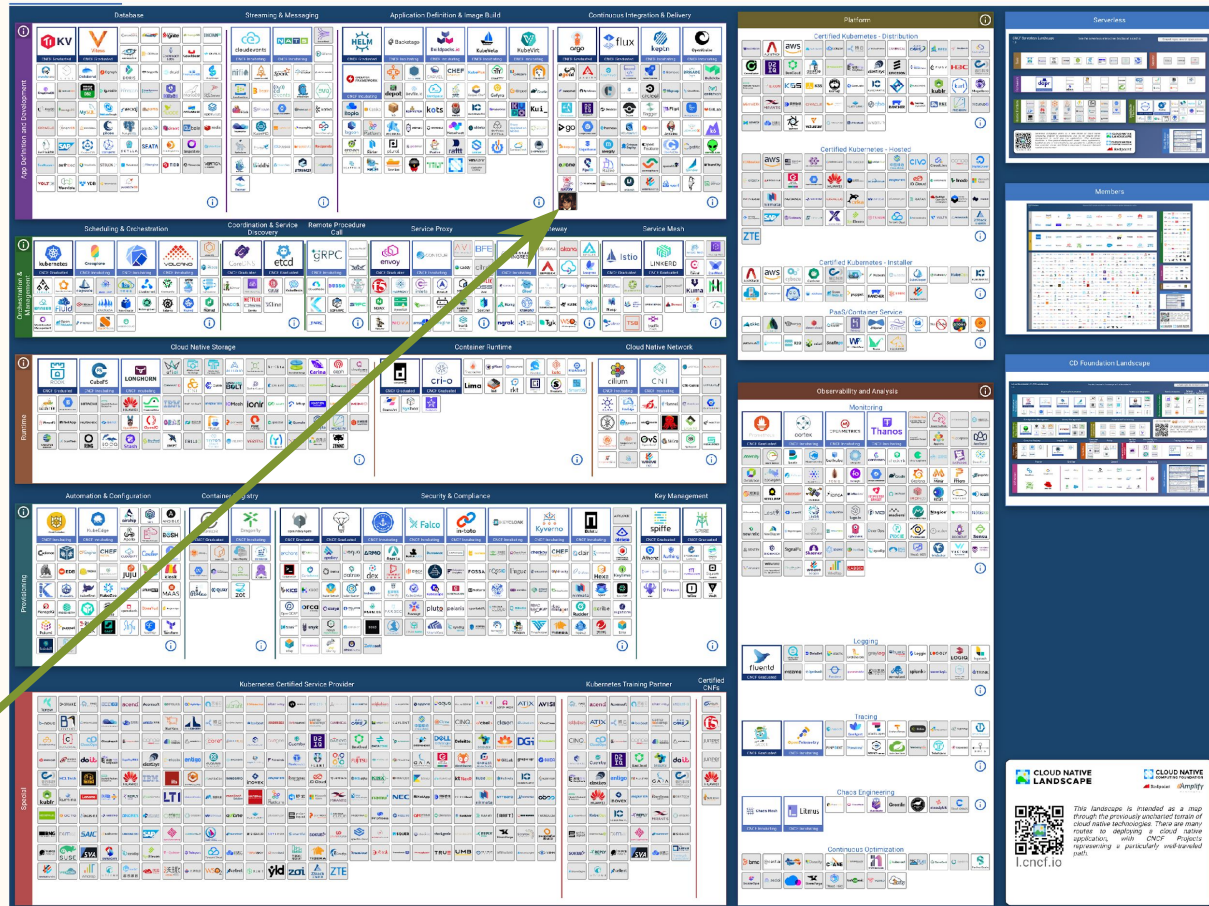
~~Where's Waldo?~~
~~Where's Charlie?~~
~~Where's Wally?~~
Where's Harry?



Source: <https://landscape.cncf.io/> - <https://nl.ign.com/harry-potter-the-series>

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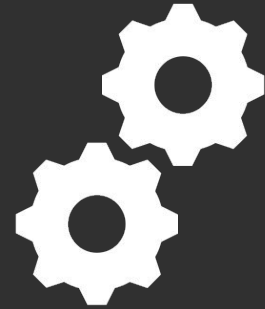


Source: <https://landscape.cncf.io/> - <https://nl.lgn.com/harry-potter-the-series>

But... How to overcome these struggles?

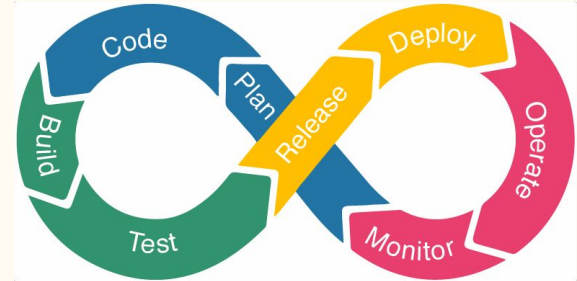


Platform Engineering

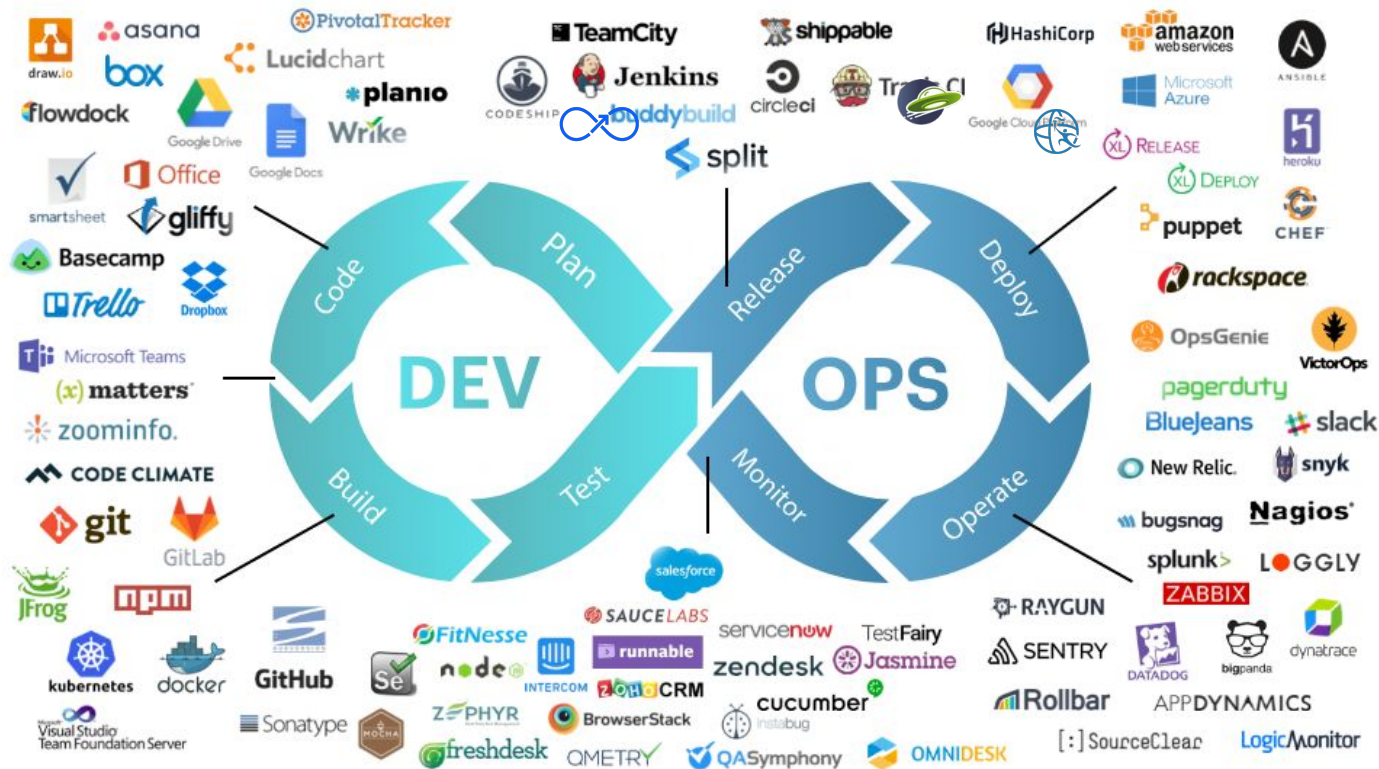


The Platform Engineering recipe - Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools

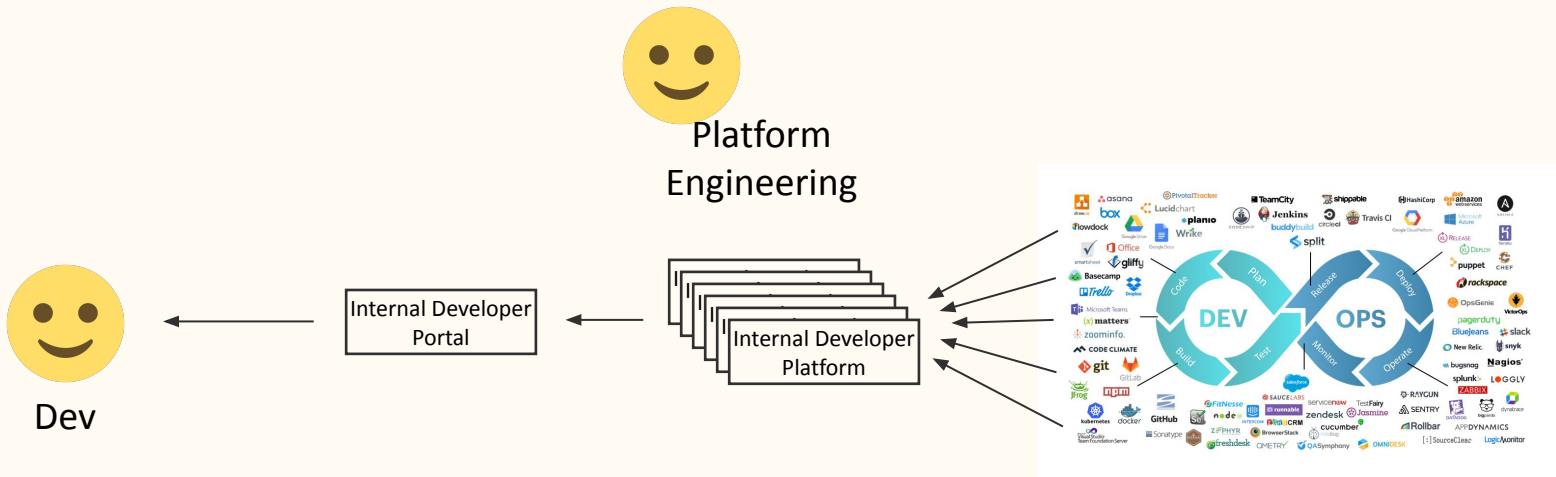


Bake your own Platform



Source: <https://www.openxcell.com/devops/>

So, what does this look like?



Organizational Impact & Cognitive Load



Dev

Applications



Platform Engineering

App Build

App Deploy

Entitlement/Billing

Runtime

Java

Guest OS



Ops

Entitlement/Billing

Guest
Virtualization

Clustering/HA

Host Virtualization

OS

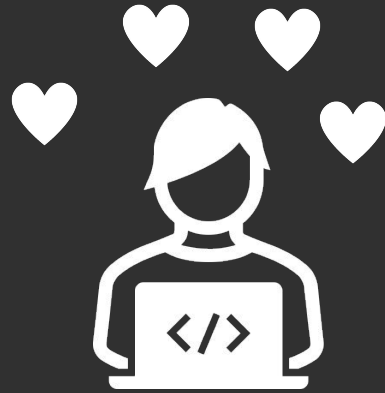
Hardware

Internal Developer Platform

Developer Portal

!! Product

*Platforms and
portals*



Demo - Backstage

Openshift - developer hub (show developer hub in one minute)
⇒ now pop up the flying letters

Flying letters:

<https://d1e4pidl3fu268.cloudfront.net/57759859-89af-4a67-9b3b-59f3e6c5cbec/letters.jpeg>



Flying letters:

<https://tenor.com/view/harry-potter-sorcerers-stone-daniel-radcliffe-mail-jump-gif-5101797>





A developer portal on OpenShift is like Hogwarts for developers: templates are your spell books, pipelines your enchanted corridors, while governance and security act like professors keeping things safe.

The portal itself, your Room of Requirement, always giving you exactly what you need (e.g., to spin up new projects) with a flick of your wand.

Keep your options open





Maarten Vandeperre

*Master of applied Java charms and
function-calling rituals.*



Camille Nigon

Professor of Magical Intelligence

Source:

https://contentful.harrypotter.com/usf1vwтуqyxm/1LwRJ7C8d6q8kmAOo4QAcc/ed6000b0abd1aac4e63ef10446ec6e2/HarryPotter_PM_B4C9M1_HarryRonHermioneSeeDarkMark_Moment.jpg?q=75&fm=jpg&w=2560

Pop-up image



Pop-up image



Pop-up image



Sovereignty (framework)

#	Sovereignty Objectives	Weight in Scoring
SOV-1	Strategic Sovereignty	15%
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SOV-8	Environmental Sustainability	5%
Total		100%

Twee problemen die ik voorzie

- Relatief korte levensduur van bepaalde modellen gaat ervoor zorgen dat we heel veel test werk gaan moeten doen wanneer we ergens in onze processen een LLM gebruiken dat op zogenaamd "pensioen" gaat.
 - Kan ik [redacted] ervoor zorgen dat deze modellen ergens autonoom kunnen blijven doordraaien? Als ze op een bepaald moment in de tijd goed genoeg waren voor onze use-case, dienen we deze niet noodzakelijk te vervangen door een ander model, ookal is dit beter
- Momenteel gebruiken we bepaalde [redacted]-services die, vanwege te grote vraag, ons limiteren in het aantal calls/s voor een specifiek LLM. Hierdoor is in de toekomst misschien mogelijk dat onze toepassingen niet de gewenste performantie hebben (vb. wanneer we voor 1 vraag veel parallele calls moeten doen naar hetzelfde taalmodel, in combinatie van meerdere gebruikers die dit op dezelfde moment wensen te doen).

Pop-up image



Source:

<https://www.usmagazine.com/wp-content/uploads/2018/08/harry-potter-inspired-cocktails.jpg?crop=182px%2C94px%2C1539px%2C869px&resize=1600%2C900&quality=86&strip=all>

Pop-up image



Source:

<https://static0.gamerantimages.com/wordpress/wp-content/uploads/2023/08/hp-snape-double-agent.jpg?q=50&fit=crop&w=1100&h=618&dpr=1.5>

Pop-up image



Source: <https://thewonderofminiatures.home.blog/wp-content/uploads/2012/11/f632e-repottingmandrakes.jpg>
Code: <https://github.com/maarten-vandeperre/cdc-based-integration-example/tree/main>

Pop-up image



Room of Requirement / Dev Portal

The Room of Requirement is a magical, secret room in Hogwarts Castle that appears only when a person is in great need of it, providing whatever the user requires.



Recap

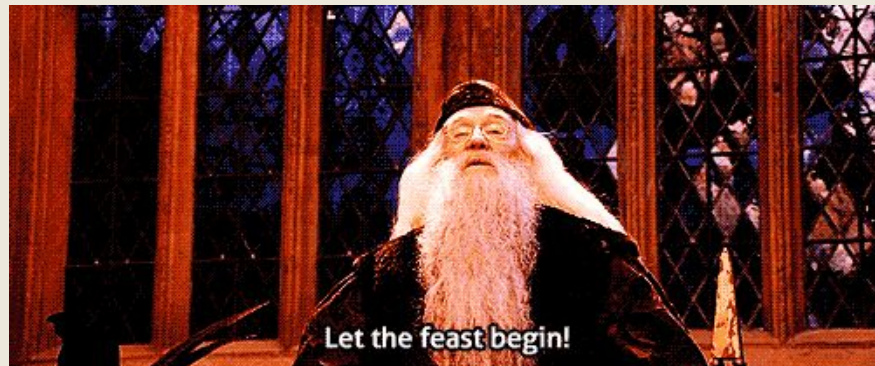
- ✓ Reduce sovereign AI's cognitive load with platform engineering
- ✓ Embed security in your CI pipelines
- ✓ Break the silos: collaborate for victory (and against the dark forces)
- ✓ Keep in control of your data and your models
- ✓ Find out how to enter your Room of Requirement
developers.redhat.com



We turn developer work into wizardry. ✨

Digital sovereignty isn't that complex...

... Stay in control!!!

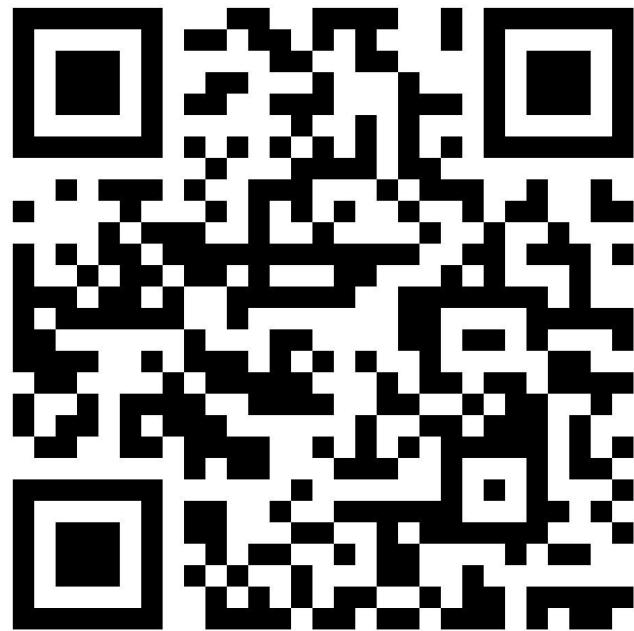
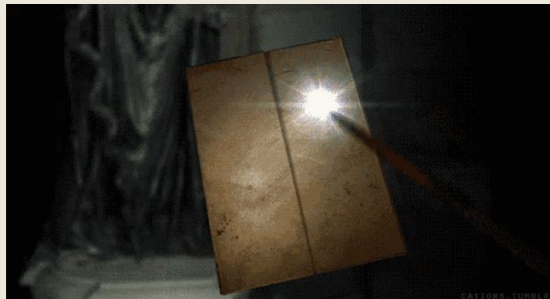


By Order of the Hogwarts Council of Developers ...

*Those who found this session “most illuminating”
often feel mysteriously drawn to tap certain symbols in the app.*



Curious thing...the Marauder's Map just lit up. Coincidence?



Thanks!



<https://www.linkedin.com/in/camille-nigon>



<https://www.linkedin.com/in/maarten-vandepierre>

<https://maarten-vandepierre.be>



<https://github.com/camille-maarten/presentations>



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Defence against the dark arts of AI

VOXXED DAYS TICINO

FEBRUARY 2026

VOL I No I

The playbook for a sovereign model-as-a-service platform



Maarten Vandeperre
Solutions Architect



Camille Nigon
Solutions Architect



Keep your options open



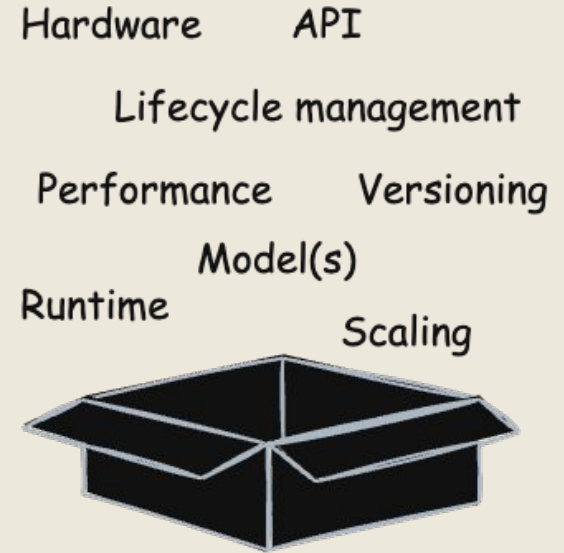
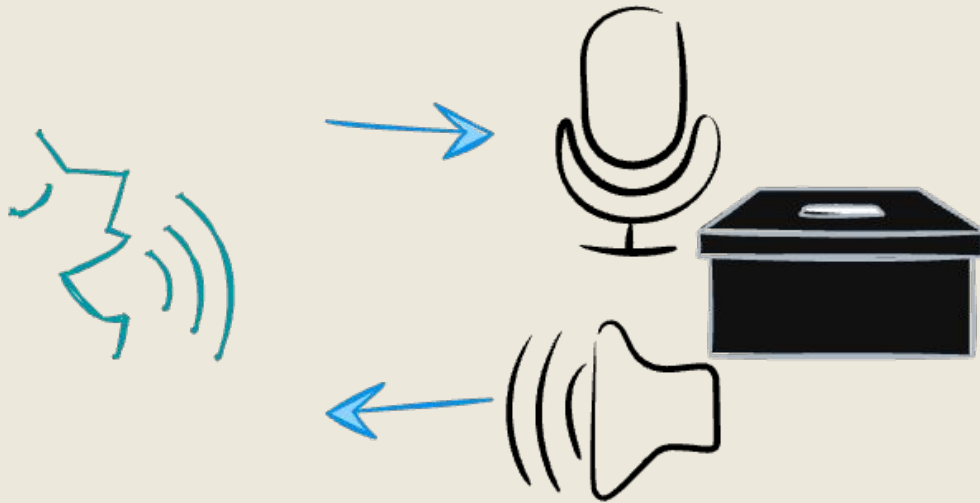
Muggle world



Show of hands



Model as a service



Model as a service

Minimalistic implementation - demo

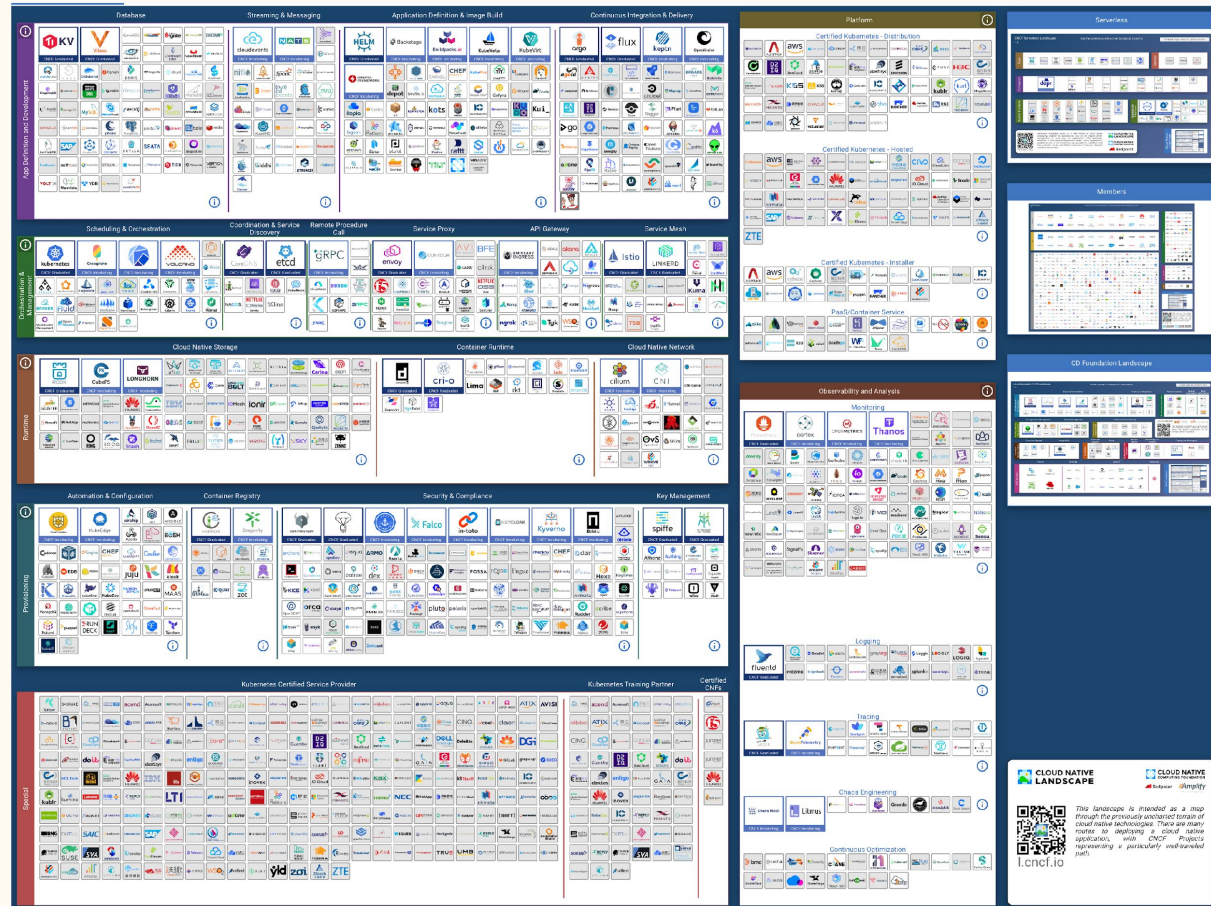
About sponsoring...

maarten-vandepierre.be



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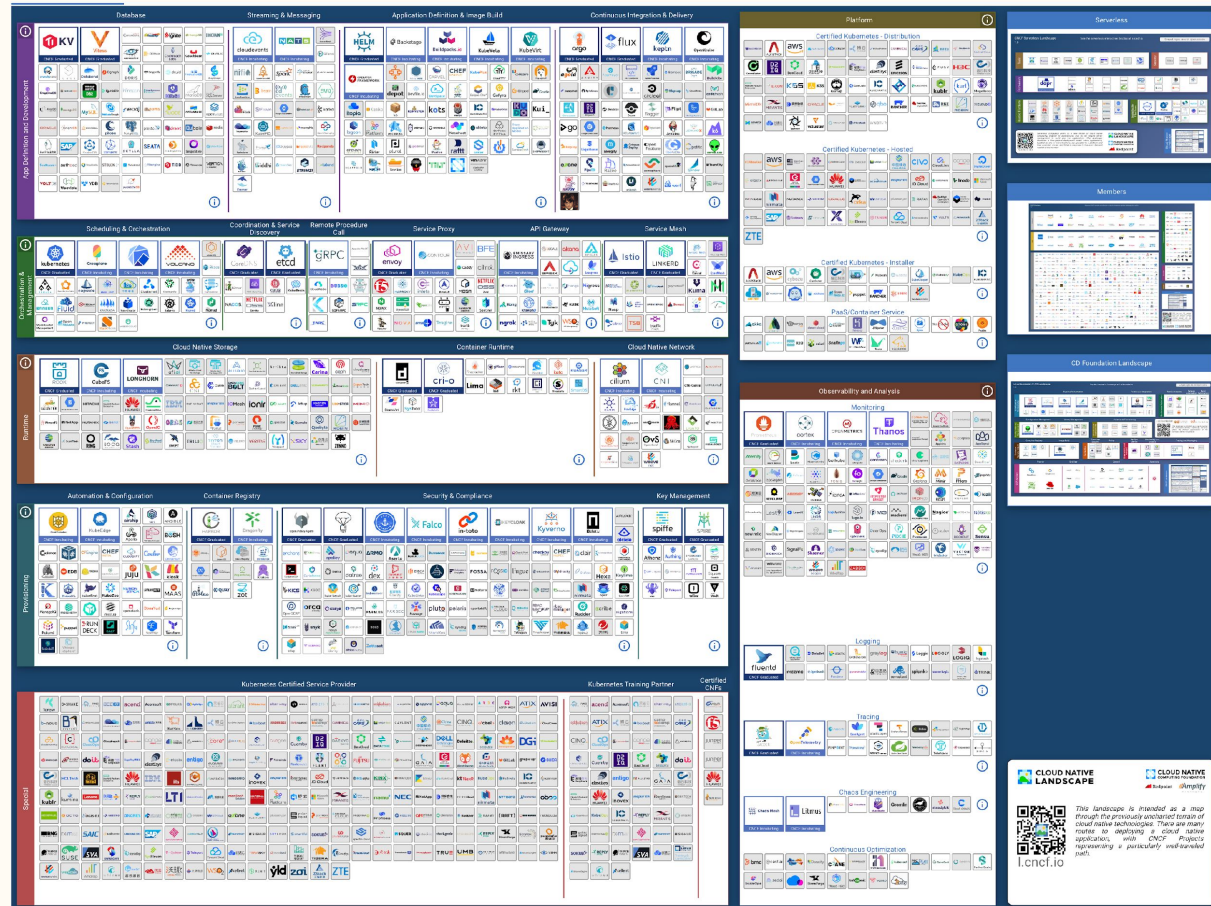
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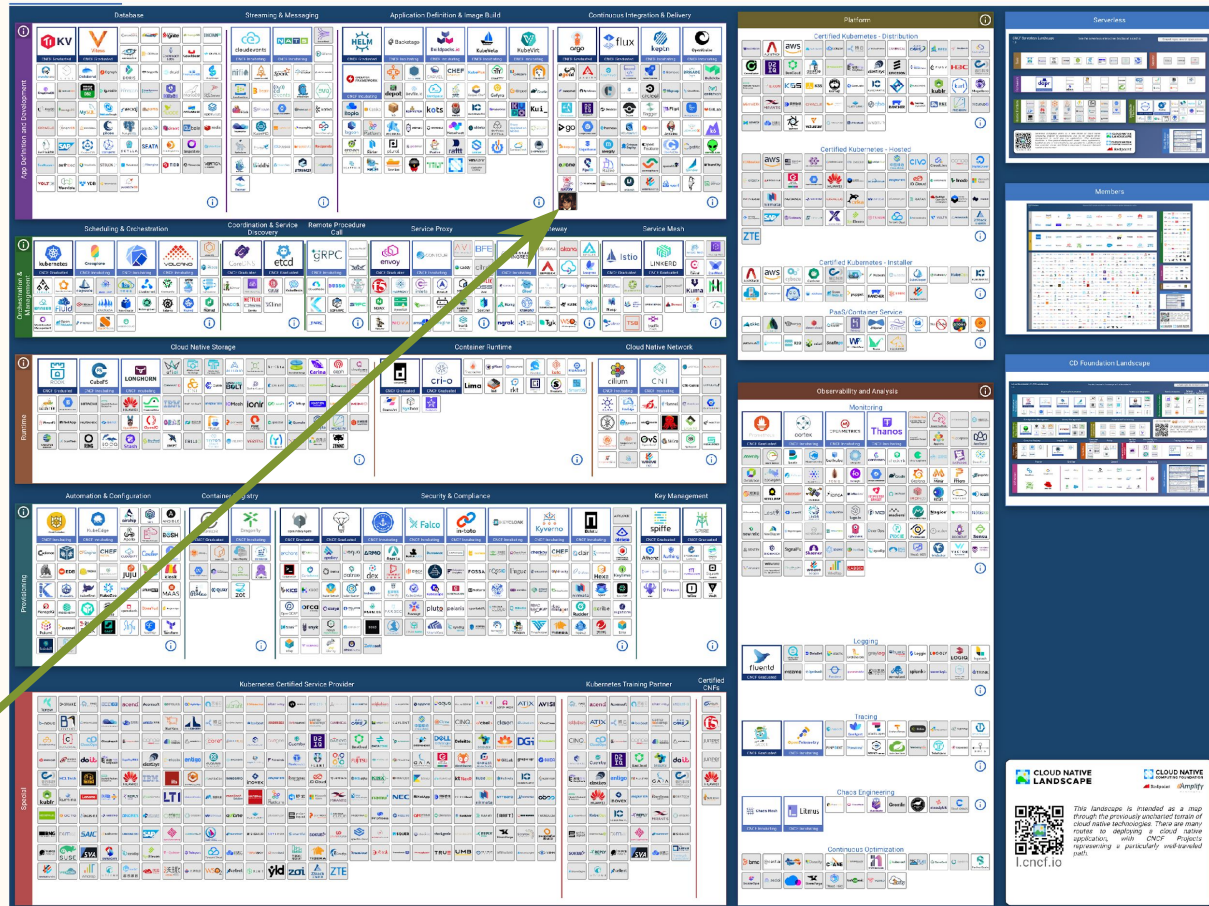
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... Now that we got our money

Back to the muggle world

What's in the news?

What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant



Wire
Jul 15, 2025



NEWSLETTERS: CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT



GETTY IMAGES

Forbes

EDITORS' PICK | BUSINESS > MANUFACTURING

Why 95% Of AI Pilots Fail, And What Business Leaders Should Do Instead

By [Andrea Hill](#), Contributor. © Andrea Hill is a multi-industry CEO covering bu...

Follow Author

Published Aug 21, 2025, 11:54am EDT, Updated Aug 28, 2025, 07:54am EDT

Source:

<https://www.forbes.com/sites/andreahill/2025/08/21/why-95-of-ai-pilots-fail-and-what-business-leaders-should-do-instead/>

<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>

Sovereignty

Sovereignty (framework)

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<https://d1e4pidl3fu268.cloudfront.net/57759859-89af-4a67-9b3b-59f3e6c5cbec/letters.jpeg>



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<https://tenor.com/view/harry-potter-sorcerers-stone-daniel-radcliffe-mail-jump-gif-5101797>





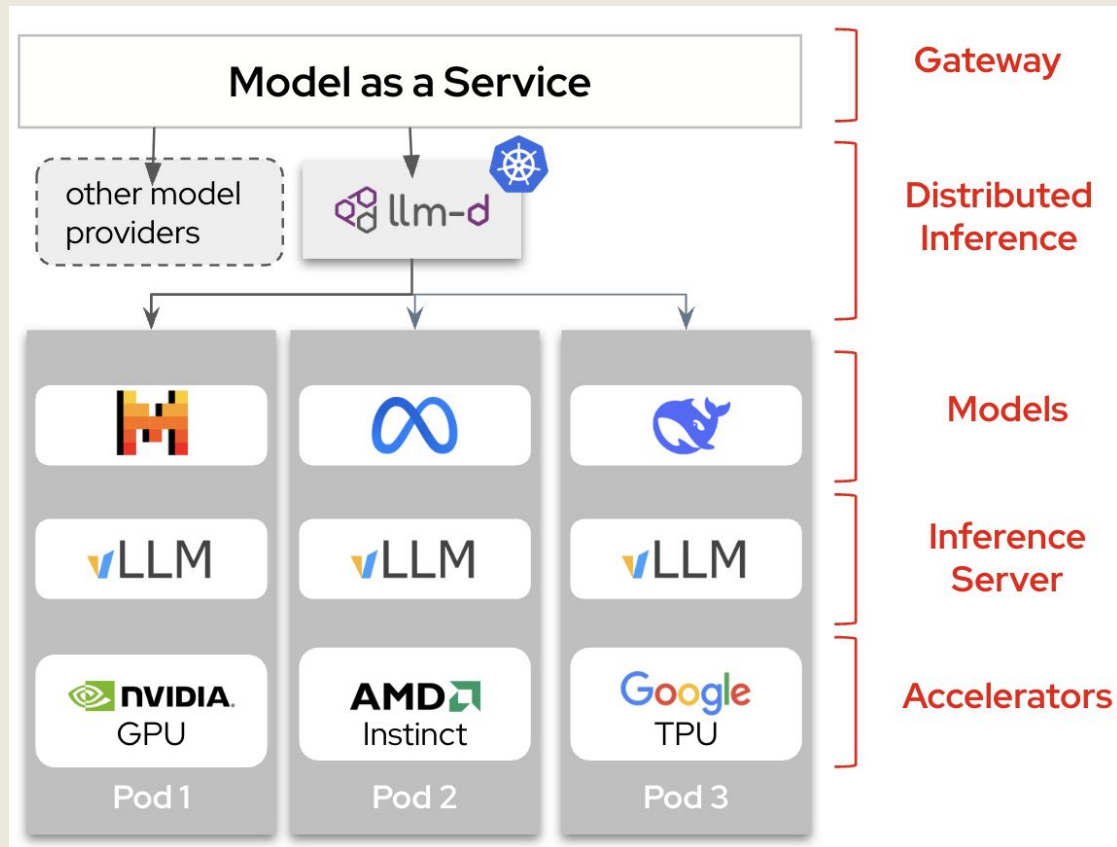
Hogwarts - sovereign platform



Platform - Room of requirements

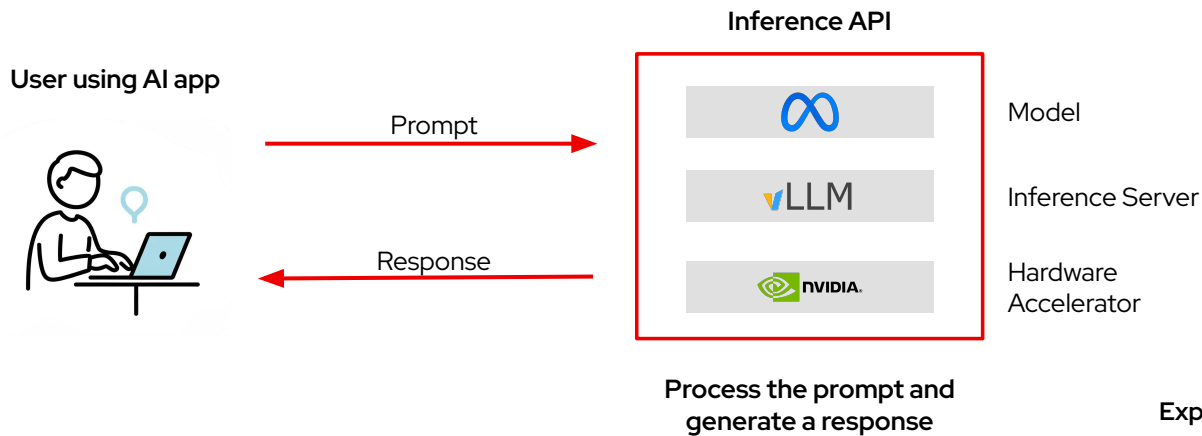


Model as a service

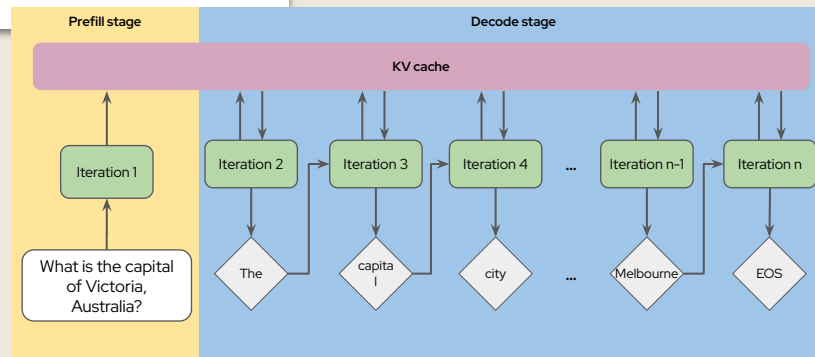


Inferencing

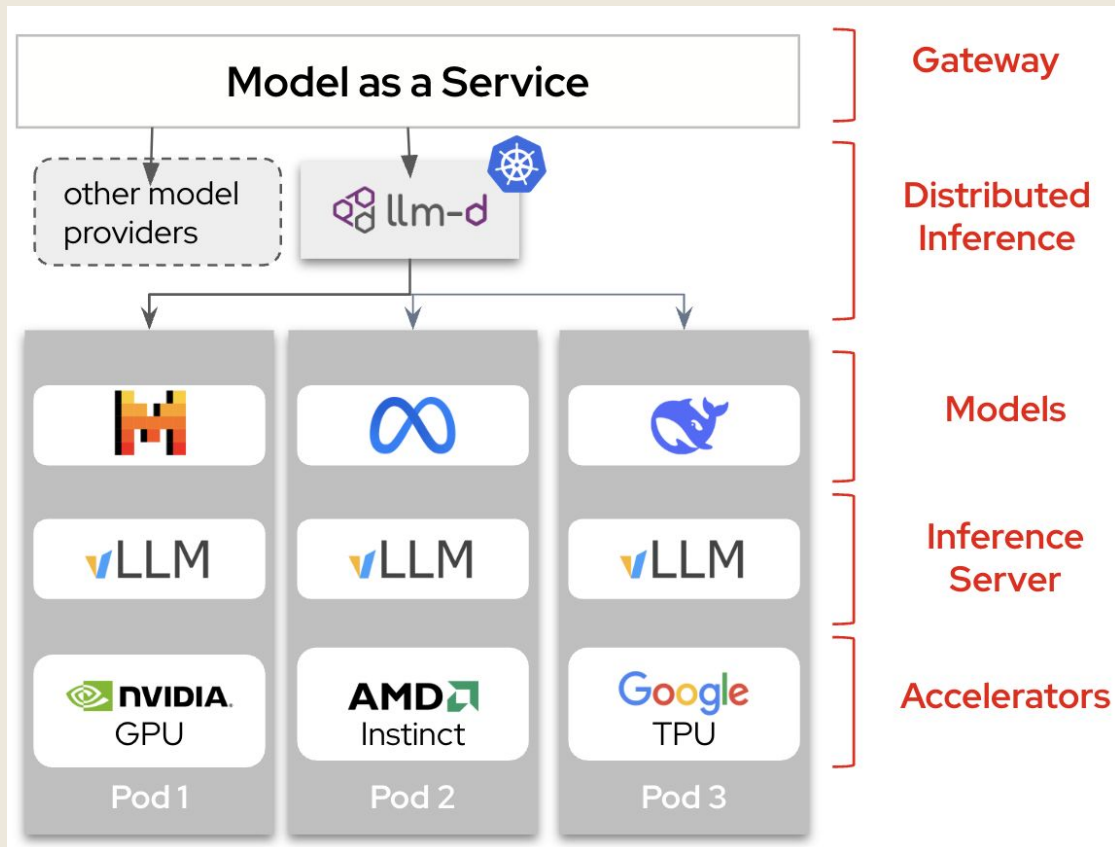
Casting the spell



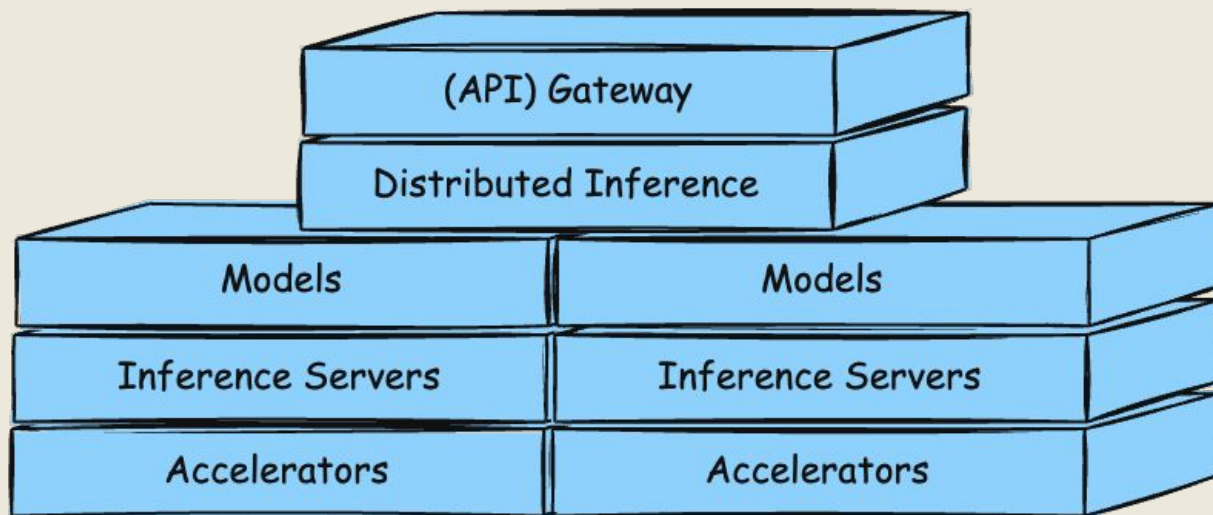
Exploiting the transformer architecture



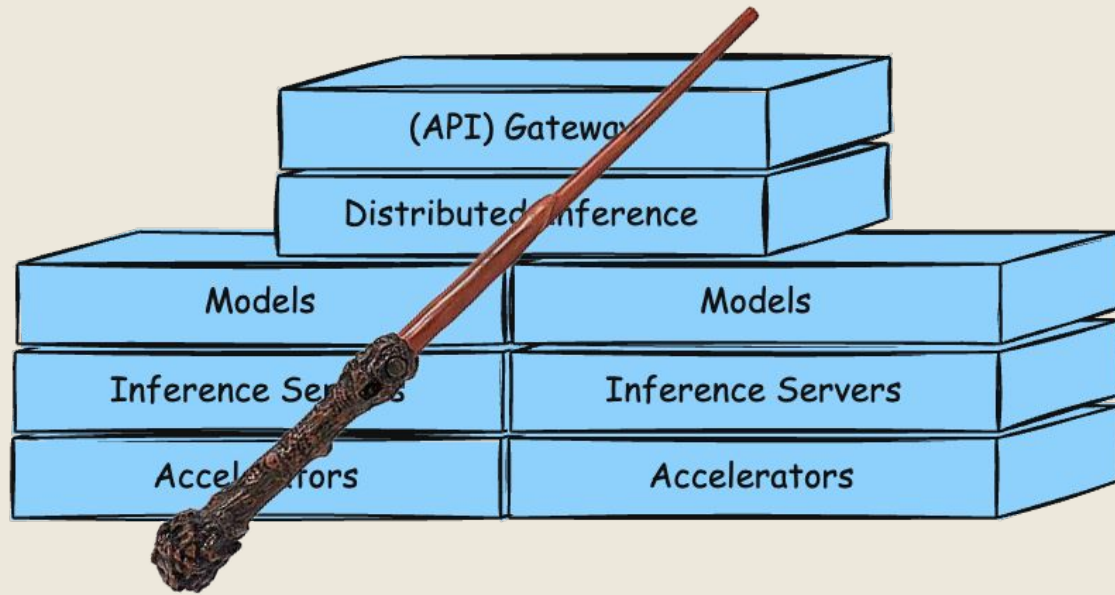
Model as a service



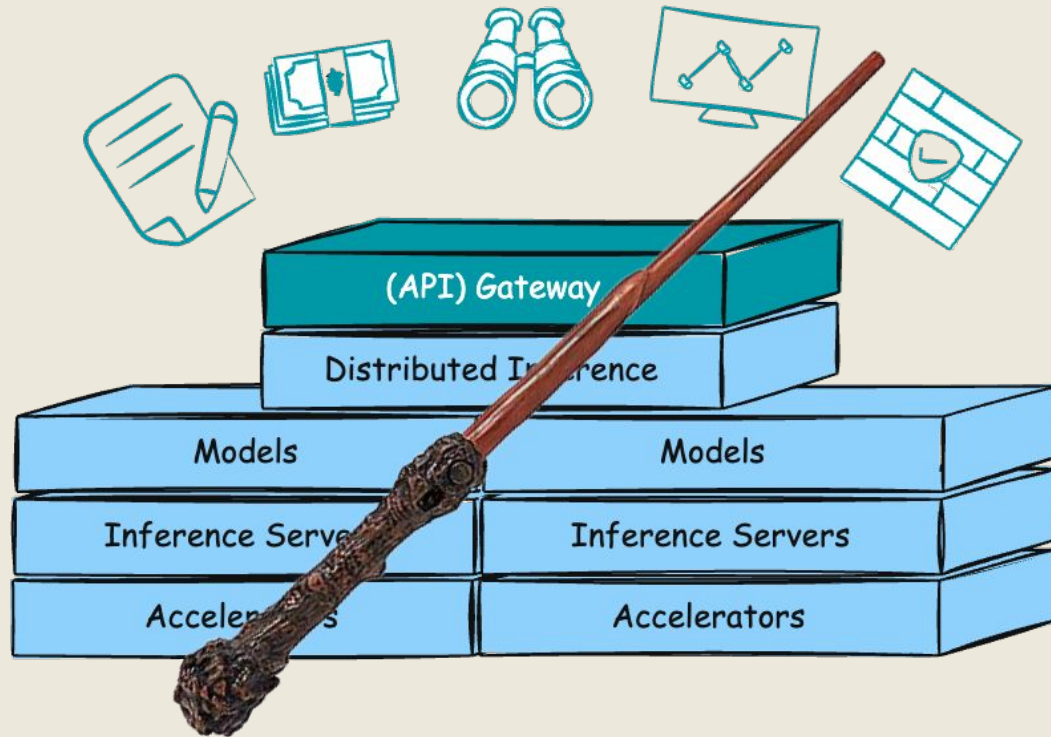
Model as a service



Model as a service



Model as a service



One want to rule them all???

Hugging Face Search models, datasets, users...

Models 2,200,944 Filter by name

Full-text search Inference Available 11 Sort: Trending

Main Tasks Libraries Languages Licenses Other

Tasks

- Text Generation
- Any-to-Any
- Image-Text-to-Text
- Image-to-Text
- Image-to-Image
- Text-to-Image
- Text-to-Video
- Text-to-Speech
- + 42

Parameters

< 1B 6B 12B 32B 128B > 500B

Libraries

- PyTorch
- TensorFlow
- JAX
- Transformers
- Diffusers
- Safetensors
- ONNX
- GGUF
- Transformers.js
- MLX
- Keras
- + 41

Apps

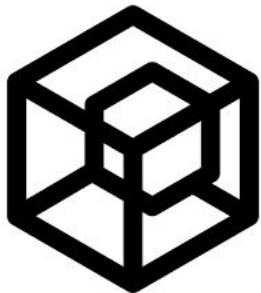
- vLLM
- TGI
- llama.cpp
- MLX LM
- LM Studio
- Ollama
- Jan
- + 13

Inference Providers

- Groq
- Novita
- Nebius AI
- Cerebras
- SambaNova
- Nscale
- fal
- Hyperbolic
- + 10

Models

- MiniMaxAI/MiniMax-M2**
Text Generation · 229B · Updated 4 days ago · 630k · 933
- moonshotai/Kimi-Linear-48B-A3B-Instruct**
Text Generation · 49B · Updated 1 day ago · 9.06k · 295
- meituan-longcat/LongCat-Video**
Text-to-Video · Updated 4 days ago · 1.04k · 258
- PaddlePaddle/PaddleOCR-VL**
Image-Text-to-Text · 1.0B · Updated 2 days ago · 26.7k · 1.19k
- openai/gpt-oss-safeguard-20b**
Text Generation · 22B · Updated 4 days ago · 4.41k · 104
- ibm-granite/granite-4.0-h-1b**
Text Generation · 1B · Updated 10 days ago · 2.65k · 91
- LiquidAI/LFM2-CoBERT-350M**
Sentence Similarity · 0.4B · Updated 5 days ago · 599 · 82
- nvidia/ChronoEdit-14B-Diffusers**
Image-to-Image · Updated 4 days ago · 350 · 72
- BAAT/Emu3.5**
34B · Updated 2 days ago · 172 · 69
- valiantcat/Qwen-Image-Edit-MeITU**
Image-to-Image · 20B · Updated 7 days ago · 10.2k · 122
- deepseek-ai/DeepSeek-OCR**
Image-Text-to-Text · 3B · Updated 9 days ago · 1.85M · 2.37k
- briaai/FIBO**
Text-to-Image · Updated about 2 hours ago · 2.46k · 170
- Soul-AILab/SoulX-Podcast-1.7B**
Text-to-Speech · 2B · Updated about 5 hours ago · 738 · 129
- datalab-to/chandra**
Image-to-Text · 9B · Updated 12 days ago · 12.6k · 178
- dx8152/Qwen-Edit-2509-Multiple-angles**
Image-to-Image · Updated about 5 hours ago · 104
- nvidia/audio-flamingo-3-hf**
Audio-Text-to-Text · 8B · Updated 6 days ago · 482 · 88
- Phz00t/Qwen-Image-Edit-Rapid-AI0**
Text-to-Image · Updated 1 day ago · 517
- lightonai/LightOnOCR-1B-1025**
Image-to-Text · Updated about 22 hours ago · 10.4k · 129
- Phz00t/WAN2.2-14B-Rapid-A11InOne**
Image-to-Video · Updated about 14 hours ago · 1.02k
- Qwen/Qwen3-VL-2B-Instruct**
Image-Text-to-Text · 2B · Updated 10 days ago · 112k · 153



Demo - wand comparisson

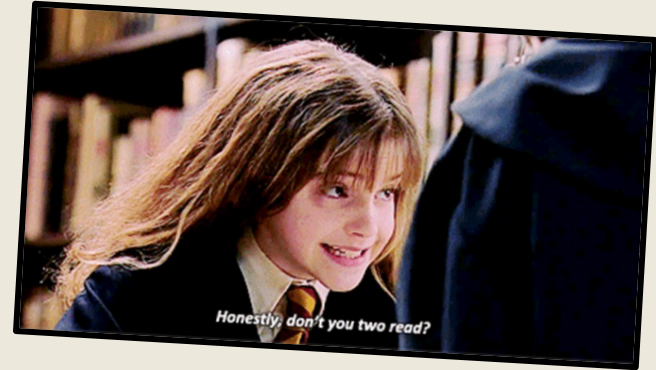
Keep your options open



*Know the context,
set the scene*

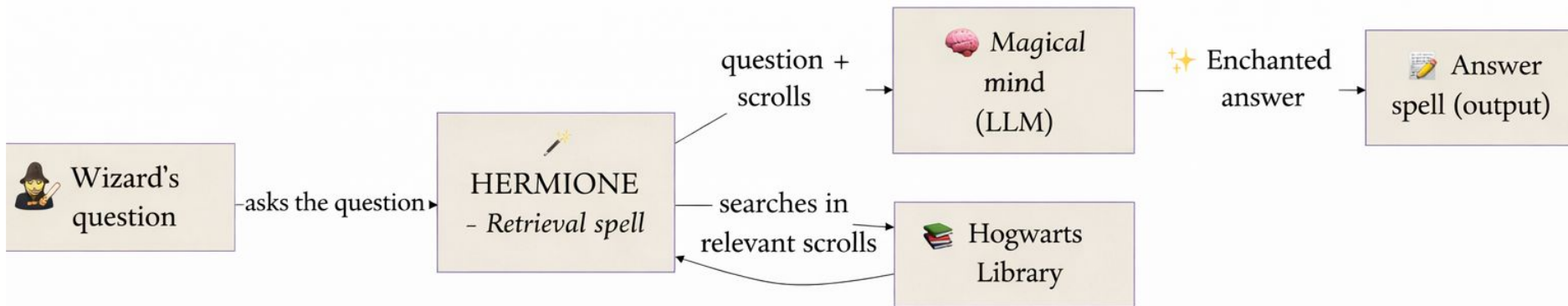
Hermione Granger

- ✓ Brilliant, methodical, and always prepared
- ✓ Specialist in books and information research

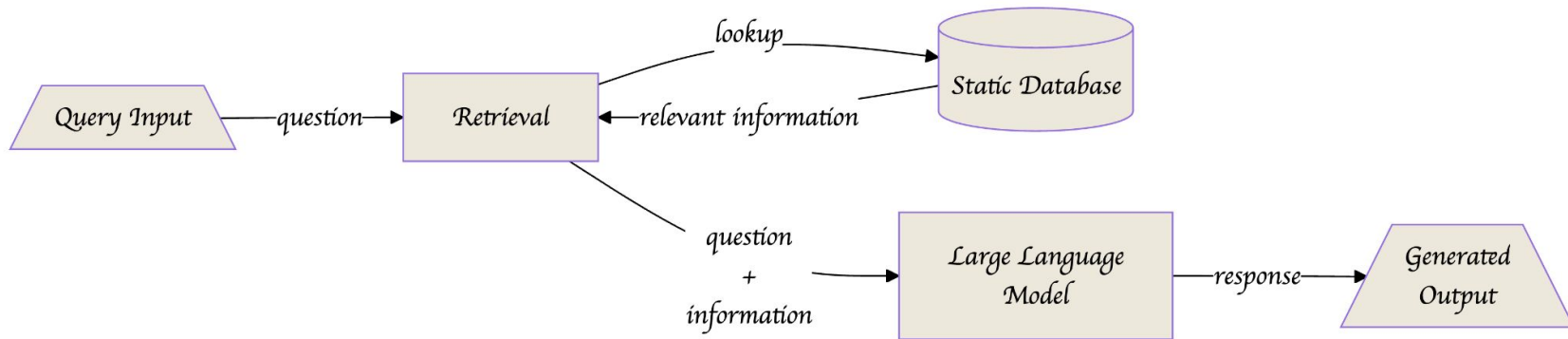


“A witch is capable of solving any problem... if she has access to the right source.”

The magic of Hermione



RAG



Ron Weasley

- ✓ A pre-trained language model that has been fine-tuned on specific tasks
- ✓ No access to an external database
- ✓ Trained to respond well in certain situations



"Once the piece is placed, you can no longer change the plan."

Ron vs. Hermione



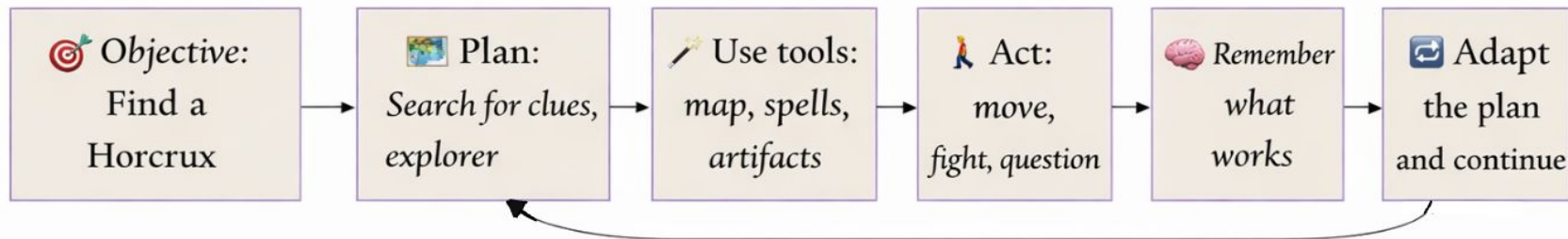
	Hermione (RAG)	Ron (Fine-tuning)
External access	Yes	No
Response time	Slower	Fast
Knowledge updates	Simple	Difficult
Adaptability	High	Medium
Inference cost	Higher	Lower* (depends on model selection)

Harry Potter

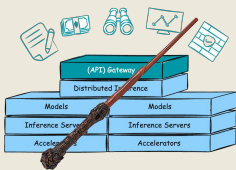
- ✓ Clear objectives (* most of the time)
- ✓ Makes his own decisions without waiting for orders
- ✓ Uses magical tools:
 - Magic wand
 - Invisibility cloak
 - Marauder's Map
- ✓ Learns from past experiences to improve
- ✓ Adapts quickly



The magic of Harry



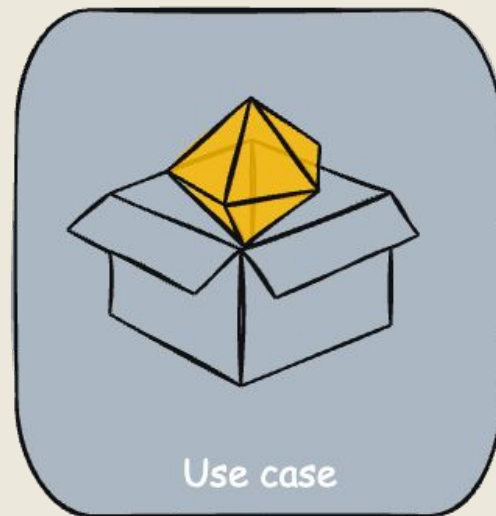
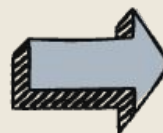
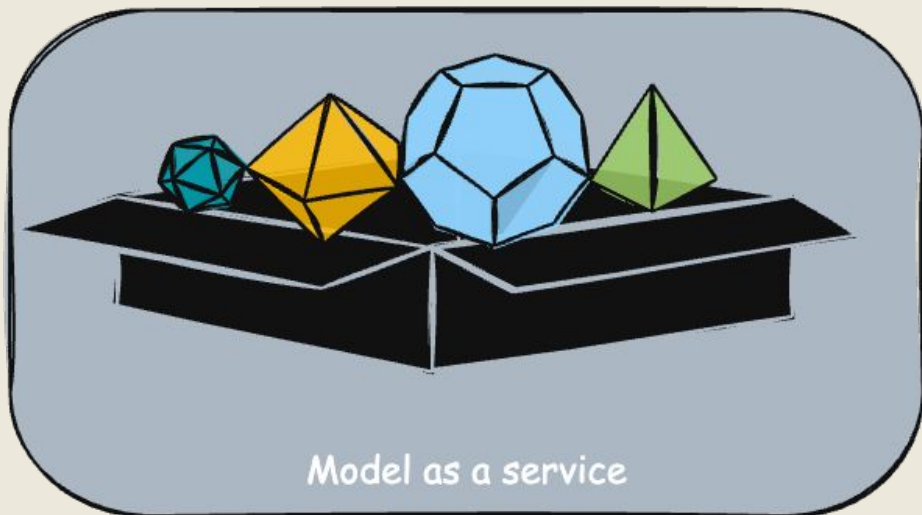
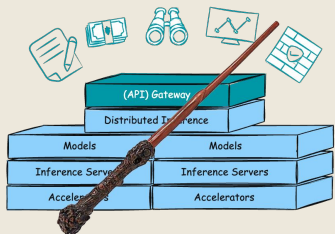
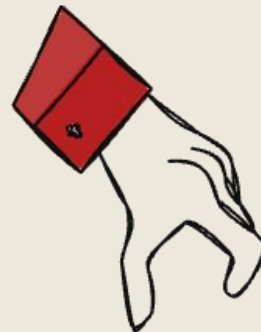
*How does this relate to
model as a service?*

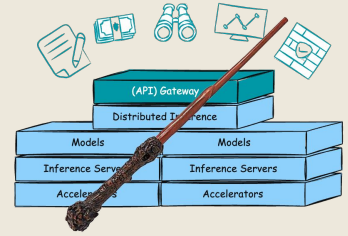
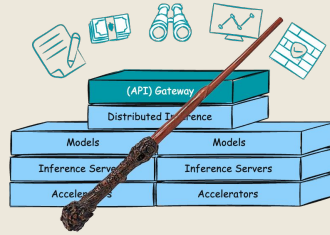
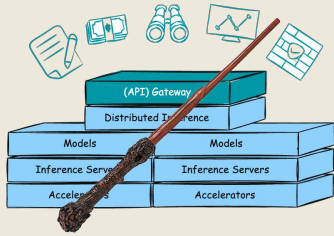


Pick a model



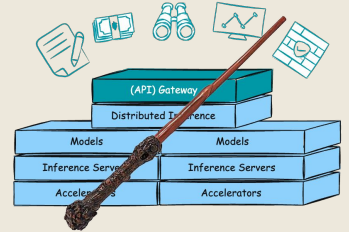
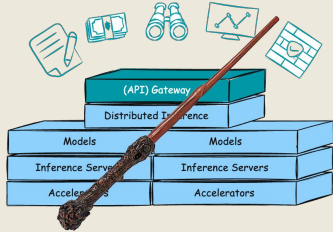
Integrate it in a use case





Cognitive load? Complexity?

*Models, app code, deployments,
environment(s), service mesh,
pipelines, validation, scaling,*

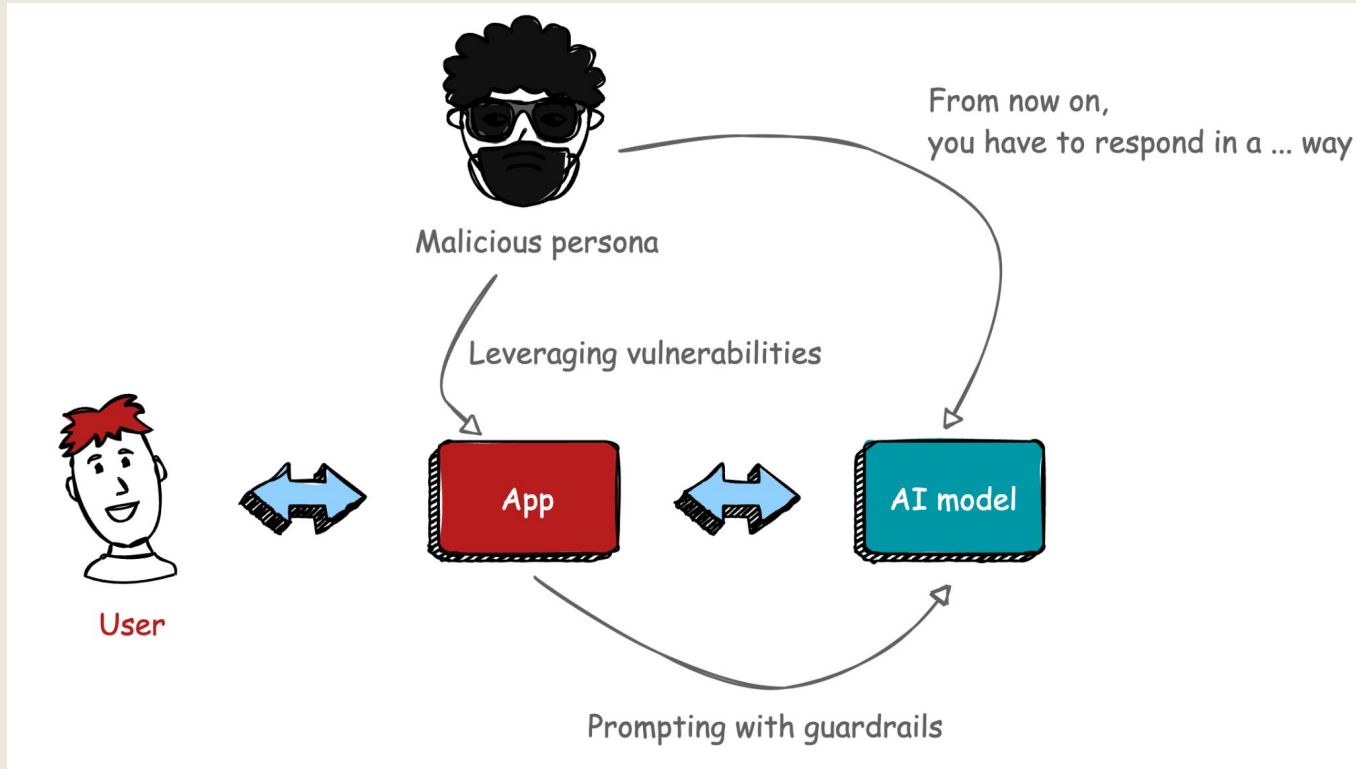


Platform engineering

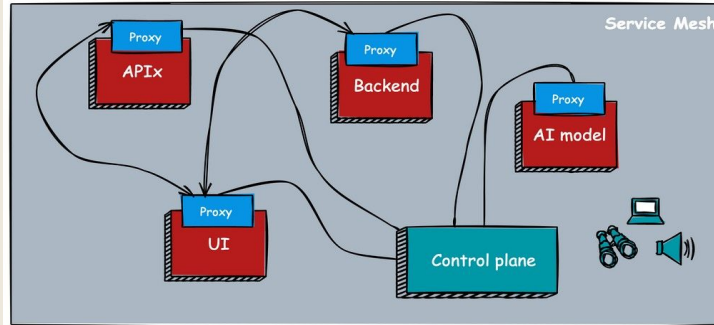
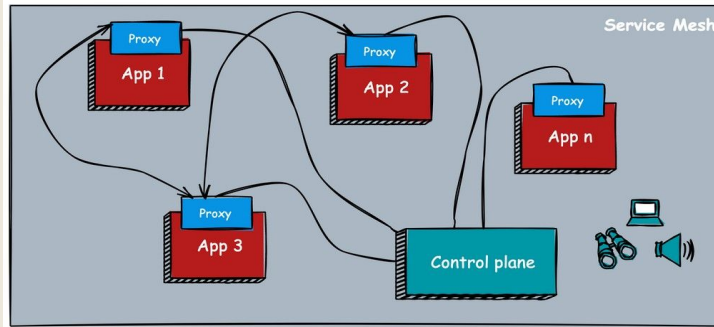
Blueprints/scaffolding templates

How do we protect our wand?

Jailbreak



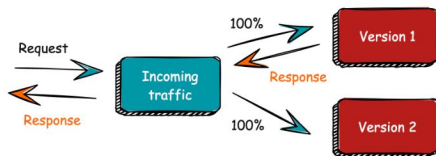
Service mesh



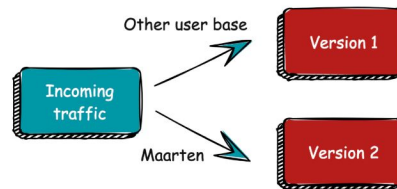
Communication: App x can communicate with App y within the service mesh

Service mesh

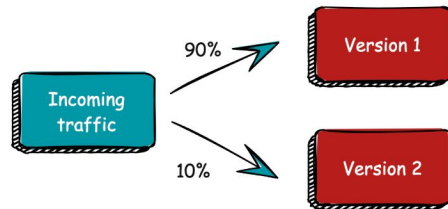
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



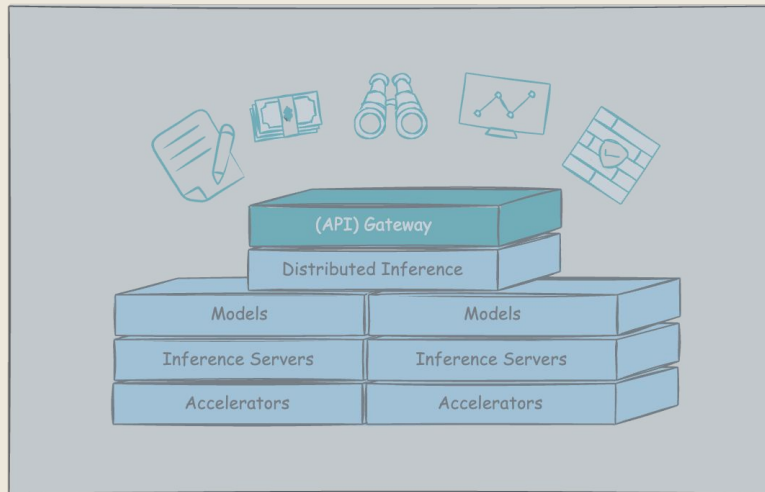
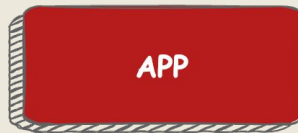
Testing - Canary releases



Rolling out - Blue/green deployments

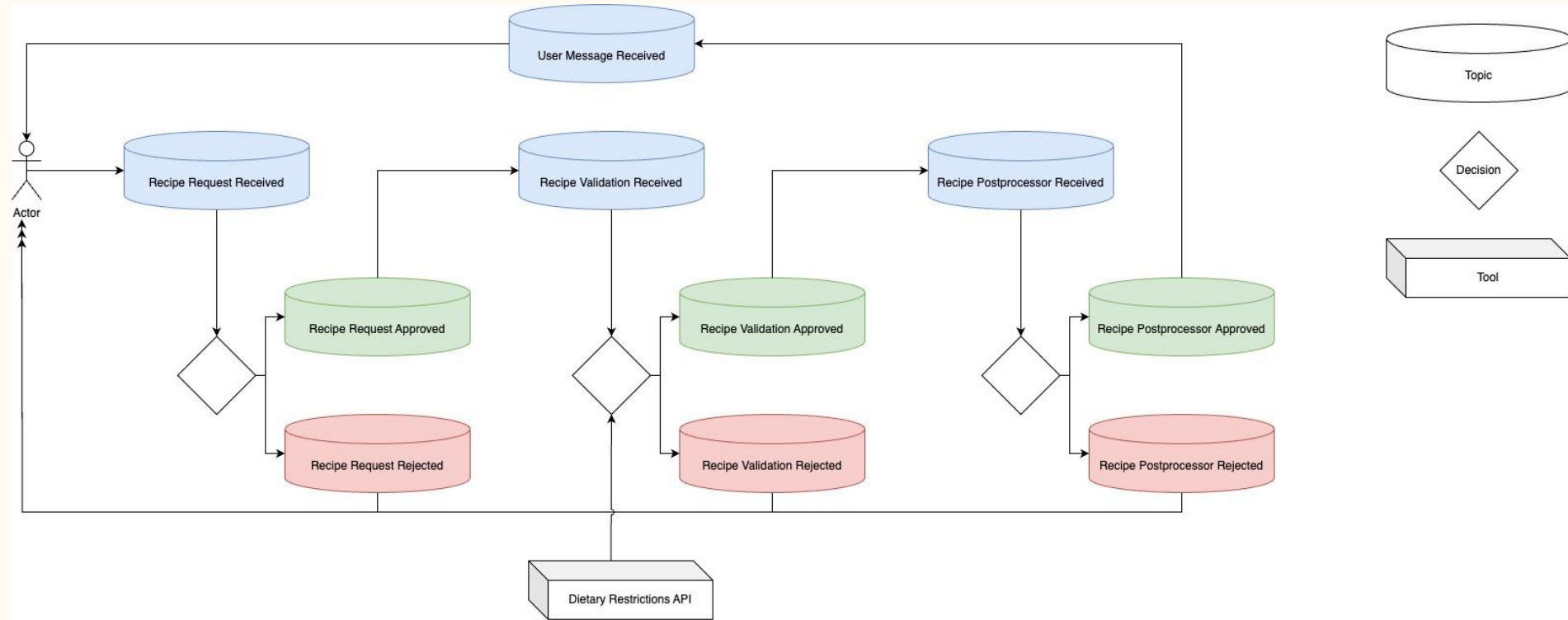
Demo - scaffolding template

Demo - scaffolding template



How do we protect our wand
ll?

Guardrails



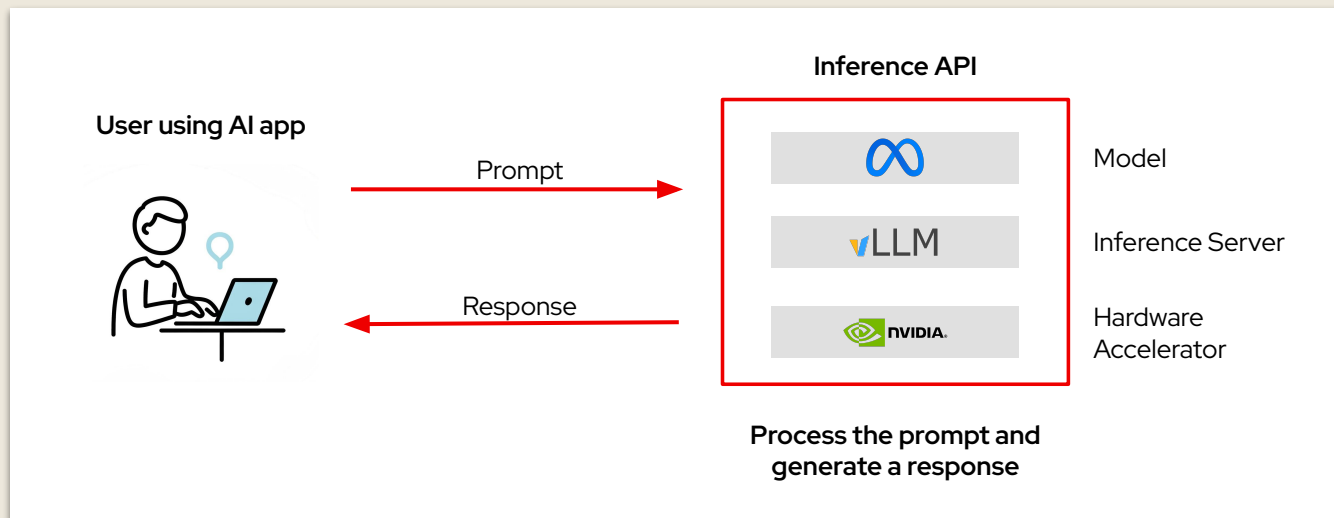
Demo video:

<https://youtube.com/watch?v=j5HuRWWi5mA&si=uFMA7EPIMPLqJHg>

Demo - service mesh

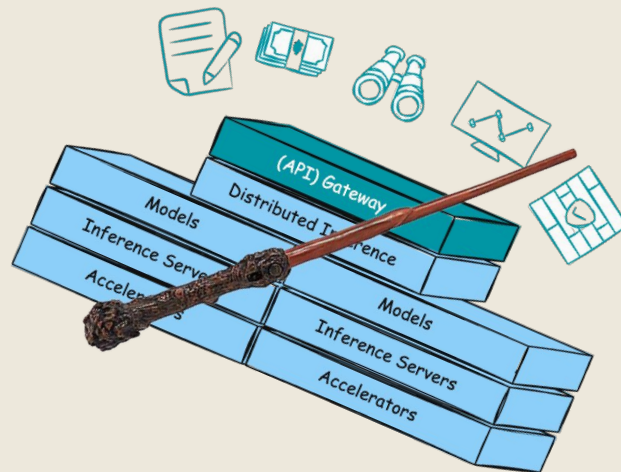
Demo - guardraíling

Demo - guarddrailing



Recap

- ✓ Think about sovereignty
- ✓ Monitor, monitor, monitor (maybe through monetization)
- ✓ Keep in control of your data and your models
- ✓ Reduce cognitive load
- ✓ Find out how to enter your Room of Requirement, how to start with MaaS
developers.redhat.com

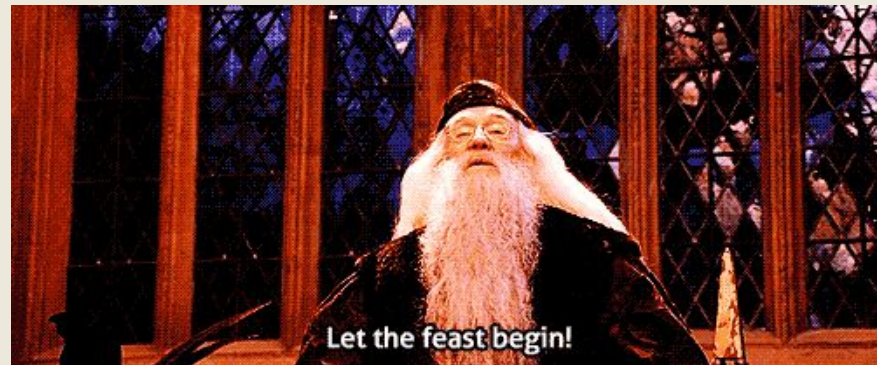


We turn developer work into wizardry. ✨

Digital sovereignty isn't that complex...

... within controlled boundaries!!!

... stay in control!!!



And...

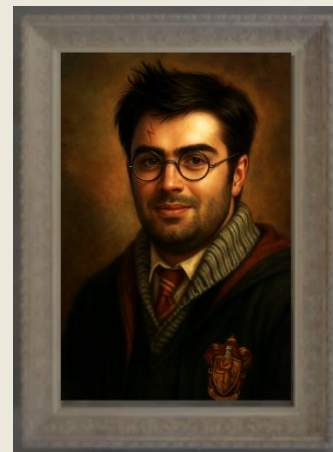
Keep your options open



Thanks!



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<https://maarten-vandepierre.be>



<https://github.com/camille-maarten/presentations>



<https://github.com/maarten-vandepierre/presentations>



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